

Project Design Phase Solution Architecture

Date	10 July 2026
Team ID	
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	4 Marks

Overview

The solution architecture bridges the manual, inconsistent loan-approval problem with an AI-powered prediction system. It defines how applicant data flows through preprocessing, the ML model, and back to the applicant as an explained decision.

Goals

- Identify the technology stack needed to automate loan approval prediction.
- Define how the Flask application, ML model, and stored artifacts interact.
- Ensure the system is maintainable and gives an explainable result, not just a label.

Architecture Layers

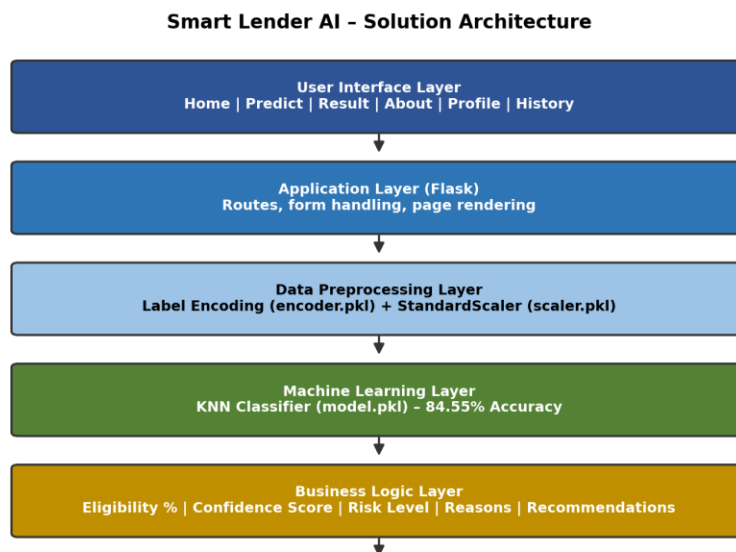


Fig 1: Smart Lender AI – Solution Architecture

Layer	Description	Technologies
1. User Interface Layer	Home, Predict, Result, About, Applicant Profile and History pages	HTML5, CSS3, JavaScript, Font Awesome
2. Application Layer	Flask routes handle form submission, validation and page rendering	Python (Flask)
3. Data Processing Layer	Label encoding of categorical fields, StandardScaler for numerical fields	Pandas, NumPy, Scikit-learn
4. Machine Learning Layer	Loads model.pkl and predicts Loan_Status; computes prediction probability	Scikit-learn (KNN), joblib

Layer	Description	Technologies
5. Business Logic Layer	Converts probability into eligibility %, confidence score, risk level, reasons, recommendations	Python (Flask application logic)

Data Flow

- Applicant submits data through the Predict page.
- Flask backend label-encodes and scales the input using encoder.pkl and scaler.pkl.
- The KNN model (model.pkl) predicts Loan_Status and its probability.
- Business logic derives eligibility percentage, confidence score, risk level, reasons and recommendations.
- The Result page displays the outcome; the record becomes available on the History and Applicant Profile pages.